

Activity: AIC vs. LRT

AIC can be used to compare any two models on the same data, with the same response variable. A likelihood ratio test, on the other hand, can only compare nested models. So, if we do have nested models, does a comparison based on AIC agree with the LRT results?

Suppose we have two nested models, a *reduced model* that is a subset of a *full model*. Recall that the LRT uses the test statistic

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

and rejects when $G > \chi^2_{q,\alpha}$, where q is the difference in number of parameters between the two models, and $\chi^2_{q,\alpha}$ is the upper α quantile of a χ^2_q distribution.

1. Show that AIC will choose the full model if $G > 2q$.
2. Suppose we use a significance level of $\alpha = 0.05$. For what values of q is the LRT more stringent than AIC for choosing the full model? (You can do this in R, you don't have to invert the χ^2 cdf by hand!)
3. If we use $\alpha = 0.01$, for what values of q is the LRT more stringent?